

System of linear Equations (線性方程組)

In mathematics, a system of linear equations (or linear system) is a collection of linear equations involving the same set of variables. For example:

$\begin{cases} 2x+3y=8 \\ 5x-y=3 \end{cases}$ is a system of two equations in the two variables *x*, *y*.

$\begin{cases} 7x-5y-7z=-8 \\ 4x-2y-3z=0 \\ 5x+4y+4z=35 \end{cases}$ is a system of three equations in the three variables *x*, *y*, *z*.

A general system of m linear equations with n unknowns can be written as

$$\begin{cases} a_{11}x_1+a_{12}x_2+\cdots+a_{1n}x_n=b_1 \\ a_{21}x_1+a_{22}x_2+\cdots+a_{2n}x_n=b_2 \\ \vdots \\ a_{m1}x_1+a_{m2}x_2+\cdots+a_{mn}x_n=b_m \end{cases}.$$

It is equivalent to a matrix equation of the form

$$A=\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \quad \mathbf{x}=\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \quad \text{and} \quad \mathbf{b}=\begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}$$

which can be written as $A\mathbf{x}=\mathbf{b}$, where *A* is called the ***coefficient matrix*** (係數矩陣).

Example $\begin{cases} 2x+3y=8 \\ 5x-y=3 \end{cases}$ may be rewritten as $\begin{pmatrix} 2 & 3 \\ 5 & -1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}=\begin{pmatrix} 8 \\ 3 \end{pmatrix}$,

with coefficient matrix $\begin{pmatrix} 2 & 3 \\ 5 & -1 \end{pmatrix}$.



Solving SLE (I): Using Inverse Matrix

Theorem: Solving System of Linear Equations by Inverse Matrix
(運用逆矩陣求線性方程組之解)

Let A be a square matrix. If A is a non-singular matrix, i.e. $\det A \neq 0$, then the system of linear equations

$$A\mathbf{x} = \mathbf{b}$$

has a unique solution given by

$$\mathbf{x} = A^{-1}\mathbf{b}$$

Example:

Use the method of inverse matrix to solve the system of equations $\begin{cases} 3x - 2y = 4 \\ x + y = 3 \end{cases}$.

Solution: The given system may be written as

$$A\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \end{pmatrix} \quad \text{where } A = \begin{pmatrix} 3 & -2 \\ 1 & 1 \end{pmatrix}.$$

Since $A^{-1} = \frac{1}{5}\begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix}$,

therefore

$$\begin{pmatrix} x \\ y \end{pmatrix} = A^{-1}\begin{pmatrix} 4 \\ 3 \end{pmatrix}$$

$$= \frac{1}{5}\begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix}\begin{pmatrix} 4 \\ 3 \end{pmatrix}$$

$$= \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

Hence the unique solution is $x = 2$ and $y = 1$.

$$\det A = \begin{vmatrix} 3 & -2 \\ 1 & 1 \end{vmatrix} = 5$$

$$\operatorname{adj} A = \begin{pmatrix} (-1)^{1+1}|1| & (-1)^{1+2}|1| \\ (-1)^{2+1}|-2| & (-1)^{2+2}|3| \end{pmatrix}^T$$

$$= \begin{pmatrix} 1 & -1 \\ -(-2) & 3 \end{pmatrix}^T$$

$$= \begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix}$$

$$A^{-1} = \frac{1}{\det A}\operatorname{adj} A$$

$$= \frac{1}{5}\begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix}$$

Not for sale



Solving SLE (I): Using Inverse Matrix

More Example:

Solve the system of equations $\begin{cases} 7x-5y-7z=-8 \\ 4x-2y-3z=0 \\ 5x+4y+4z=35 \end{cases}$.

Solution: The given system may be written as

$$A\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -8 \\ 0 \\ 35 \end{pmatrix} \quad \text{where} \quad A = \begin{pmatrix} 7 & -5 & -7 \\ 4 & -2 & -3 \\ 5 & 4 & 4 \end{pmatrix}.$$

Since $A^{-1} = \begin{pmatrix} 4 & -8 & 1 \\ -31 & 63 & -7 \\ 26 & -53 & 6 \end{pmatrix}$,

therefore

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = A^{-1}\begin{pmatrix} -8 \\ 0 \\ 35 \end{pmatrix}$$

$$= \begin{pmatrix} 4 & -8 & 1 \\ -31 & 63 & -7 \\ 26 & -53 & 6 \end{pmatrix}\begin{pmatrix} -8 \\ 0 \\ 35 \end{pmatrix}$$

$$= \begin{pmatrix} 3 \\ 3 \\ 2 \end{pmatrix}$$

Hence the unique solution is
$x=3, \quad y=3 \quad \text{and} \quad z=2$.

$$\det A = \begin{vmatrix} 7 & -5 & -7 \\ 4 & -2 & -3 \\ 5 & 4 & 4 \end{vmatrix}$$
$$= (-56+75-112) - (70-80-84)$$
$$= 1$$

$$\text{adj } A = \begin{pmatrix} \begin{vmatrix} -2 & -3 \\ 4 & 4 \end{vmatrix} & -\begin{vmatrix} 4 & -3 \\ 5 & 4 \end{vmatrix} & \begin{vmatrix} 4 & -2 \\ 5 & 4 \end{vmatrix} \\ -\begin{vmatrix} -5 & -7 \\ 4 & 4 \end{vmatrix} & \begin{vmatrix} 7 & -7 \\ 5 & 4 \end{vmatrix} & -\begin{vmatrix} 7 & -5 \\ 5 & 4 \end{vmatrix} \\ \begin{vmatrix} -5 & -7 \\ -2 & -3 \end{vmatrix} & -\begin{vmatrix} 7 & -7 \\ 4 & -3 \end{vmatrix} & \begin{vmatrix} 7 & -5 \\ 4 & -2 \end{vmatrix} \end{pmatrix}^T$$
$$= \begin{pmatrix} 4 & -31 & 26 \\ -8 & 63 & -53 \\ 1 & -7 & 6 \end{pmatrix}^T$$
$$= \begin{pmatrix} 4 & -8 & 1 \\ -31 & 63 & -7 \\ 26 & -53 & 6 \end{pmatrix}$$

$$A^{-1} = \frac{1}{\det A}\text{adj } A = \begin{pmatrix} 4 & -8 & 1 \\ -31 & 63 & -7 \\ 26 & -53 & 6 \end{pmatrix}$$

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Number of solutions in SLE

Case 1: **Consistent (有解) with unique solution (唯一解)**

$\begin{cases} 7x-5y-7z=-8 \\ 4x-2y-3z=0 \\ 5x+4y+4z=35 \end{cases}$ (shown on P.3) has a unique solution

"$x=3,\quad y=3$ and $z=2$".

Case 2: **Consistent (有解) with infinitely many solutions (無限多的解)**

Using $\begin{cases} x+2y+3z=4 \\ 2x+4y+6z=8 \\ x+y+z=6 \end{cases}$ as example:

The first equation $x+2y+3z=4$ and the second equation $2x+4y+6z=8$ are equivalent, so the SLE can be reduced to $\begin{cases} x+2y+3z=4 \\ x+y+z=6 \end{cases}$, which consists of 3 unknowns in 2 equations. As a result, there are infinitely many solutions.

Case 3: **Inconsistent / no solutions (無解)**

Using $\begin{cases} x+2y+3z=4 \\ 2x+4y+6z=5 \\ x+y+z=6 \end{cases}$ as example:

As the first equation $x+2y+3z=4$ and the second equation $2x+4y+6z=5$ are contradict with each other (自相矛盾), so there are no solutions.

Discussion:

Try to pay attention to Case 2 and Case 3. What are the characteristics of the coefficient matrices if the SLE does not have unique solution?

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Number of solutions in SLE

Theorem: Number of solutions in System of Linear Equations
(線性方程組解之數量)

Consider the system of linear equations $A\mathbf{x}=\mathbf{b}$:

If A is non-singular ($\det A\neq 0$), then the system has **unique solution (唯一解)**.
If A is singular ($\det A=0$), then the system
1. may have **no solution (無解)**, or
2. may have **infinitely many solutions (無限多的解)**.

Example:
Consider the system of linear equations

$$(S):\quad \begin{cases} x + \lambda y + z = k \\ \lambda x - y + z = 1 \\ 3x + y + 2z = -1 \end{cases} \quad \text{where } \lambda, k \in \mathbf{R}.$$

Find the values of λ such that (S) has a unique solution.

Solution: Consider

$$\begin{vmatrix} 1 & \lambda & 1 \\ \lambda & -1 & 1 \\ 3 & 1 & 2 \end{vmatrix} = (-2+3\lambda+\lambda)-(-3+2\lambda^2+1)$$
$$= -2\lambda^2+4\lambda$$
$$= -2\lambda(\lambda-2)$$

The system (S) has unique solution if and only if

$$\begin{vmatrix} 1 & \lambda & 1 \\ \lambda & -1 & 1 \\ 3 & 1 & 2 \end{vmatrix} \neq 0$$

i.e. $-2\lambda(\lambda-2)\neq 0$

$\lambda\neq 0$ and $\lambda\neq 2$

Therefore λ can be any real values except 0 and 2.

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Number of solutions in SLE

[Pure Math 1998 Paper I Q.1(a)]

Consider the system of linear equations

$$(*)\quad \begin{cases} 2x + y + 2z = 0 \\ x + (k+1)z = 0 \\ kx - y + 4z = 0 \end{cases}$$

Suppose (*) has infinitely many solutions. Find k. ***(3 marks)***

Cramer's rule (克萊瑪法則)



In the 17th century, mathematician Gabriel Cramer (加百列・克萊瑪) discovered that using ***determinants*** could calculate the ***unique solution*** of a system of linear equations. This method is faster and easier compared to solving the system of linear equations using inverse matrices.

Theorem: Cramer's Rule (克萊瑪法則) for 2 × 2 System

If the following system of linear equations has unique solution:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} e \\ f \end{bmatrix}$$

Then the solution is:

$$x = \frac{\begin{vmatrix} e & b \\ f & d \end{vmatrix}}{\begin{vmatrix} a & b \\ c & d \end{vmatrix}} = \frac{\Delta x}{\Delta} \quad \text{and} \quad y = \frac{\begin{vmatrix} a & e \\ c & f \end{vmatrix}}{\begin{vmatrix} a & b \\ c & d \end{vmatrix}} = \frac{\Delta y}{\Delta}$$

Example:

Solve the system of linear equations $\begin{cases} 5x + 2y = 13 \\ 2x - 3y = 9 \end{cases}$ by Cramer's Rule:

Solution:

$$\Delta = \begin{vmatrix} 5 & 2 \\ 2 & -3 \end{vmatrix} = -19, \quad \Delta x = \begin{vmatrix} 13 & 2 \\ 9 & -3 \end{vmatrix} = -57, \quad \Delta y = \begin{vmatrix} 5 & 13 \\ 2 & 9 \end{vmatrix} = 19$$

$\therefore$ The solution is given by

$$x = \frac{\Delta x}{\Delta} = 3 \quad \text{and} \quad y = \frac{\Delta y}{\Delta} = -1.$$

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Cramer's rule

Theorem: Cramer's Rule (克萊瑪法則) for 3 × 3 System

If the following system of linear equations has unique solution:

$$\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}.$$

Then the solution is:

$$x = \frac{\begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}}, \quad y = \frac{\begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}}, \text{ and } z = \frac{\begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}}.$$

which can be represented as $x = \frac{\Delta x}{\Delta}$, $y = \frac{\Delta y}{\Delta}$ and $z = \frac{\Delta z}{\Delta}$.

Example:

Solve the system of linear equations $\begin{cases} 7x - 5y - 7z = -8 \\ 4x - 2y - 3z = 0 \\ 5x + 4y + 4z = 35 \end{cases}$ by Cramer's Rule:

Solution:

$$\Delta = \begin{vmatrix} 7 & -5 & -7 \\ 4 & -2 & -3 \\ 5 & 4 & 4 \end{vmatrix} = 1 \quad \text{and}$$

$$\Delta x = \begin{vmatrix} -8 & -5 & -7 \\ 0 & -2 & -3 \\ 35 & 4 & 4 \end{vmatrix} = 3, \quad \Delta y = \begin{vmatrix} 7 & -8 & -7 \\ 4 & 0 & -3 \\ 5 & 35 & 4 \end{vmatrix} = 3, \quad \Delta z = \begin{vmatrix} 7 & -5 & -8 \\ 4 & -2 & 0 \\ 5 & 4 & 35 \end{vmatrix} = 2$$

$\therefore$ The solution is given by

$$x = \frac{\Delta x}{\Delta} = 3, \quad y = \frac{\Delta y}{\Delta} = 3 \quad \text{and} \quad z = \frac{\Delta z}{\Delta} = 2.$$

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Cramer's rule

[M2 2017 Q.5]

Consider the system of linear equations in real variables x, y, z

$$(E): \begin{cases} x & +2y & -z & = 11 \\ 3x & +8y & -11z & = 49 \\ 2x & +3y & +hz & = k \end{cases}, \text{ where } h, k \in \mathbf{R} .$$

Assume that (E) has a unique solution.

(i) Find the range of values of h.

(ii) Express z in terms of h and k. ***(6 marks)***

Gaussian Elimination (高斯消去法)



In fact, SLE can also be solved using the elimination method (learnt in junior forms). The elimination method appeared as early as 150 BC in the Chinese mathematical text "Nine Chapters on the Mathematical Art (九章算術)" Later, it was further developed and popularized by the mathematician Carl Friedrich Gauss (高斯) in the 17th century.

e.g.1

$$\begin{cases} x + z = 1 \\ y + z = 1 \\ x - z = 0 \\ x + y + 2z = 2 \end{cases} \Rightarrow \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & -1 \\ 1 & 1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 2 \end{pmatrix}$$

e.g.2

$$\begin{cases} x - z = 0 \\ x + y + 2z = 2 \end{cases} \Rightarrow \begin{pmatrix} 1 & 0 & -1 \\ 1 & 1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$$

The following transformations in SLE shows how the Gaussian Elimination works to eliminate one unknown after another (where R1 represents the first equation, and so on):



Standard steps in Gaussian Elimination:

1. Swap / interchange two equations:

$$\begin{cases} 3x+2y+z=13 \\ 2x+2y-2z=4 \\ 5x+y+2z=20 \end{cases} \xrightarrow{R_1 \leftrightarrow R_2} \begin{cases} 2x+2y-2z=4 \\ 3x+2y+z=13 \\ 5x+y+2z=20 \end{cases}$$

2. Multiply an equation by a non-zero constant:

$$\begin{cases} 2x+2y-2z=4 \\ 3x+2y+z=13 \\ 5x+y+2z=20 \end{cases} \xrightarrow{0.5 \times R_1 \to R_1} \begin{cases} x+y-z=2 \\ 3x+2y+z=13 \\ 5x+y+2z=20 \end{cases}$$

3. Addition of a scalar multiple of any equation to another equation:

$$\begin{cases} x+y-z=2 \\ 3x+2y+z=13 \\ 5x+y+2z=20 \end{cases} \xrightarrow[R_3-5R_1 \to R_3]{R_2-3R_1 \to R_2} \begin{cases} x+y-z=2 \\ -y+4z=7 \\ -4y+7z=10 \end{cases}$$

Not for sale

Gaussian Elimination

After some steps, one of the unknowns x has been eliminated in equations R_2 and R_3:

$$\begin{cases} 3x+2y+z=13 \\ 2x+2y-2z=4 \\ 5x+y+2z=20 \end{cases} \longrightarrow \begin{cases} x+y-z=2 \\ -y+4z=7 \\ -4y+7z=10 \end{cases}$$

Proceed further and the unknown y will be eliminated also, so we can solve z:

$$\begin{cases} x+y-z=2 \\ -y+4z=7 \\ -4y+7z=10 \end{cases} \xrightarrow{(-1)\times R_2 \to R_2} \begin{cases} x+y-z=2 \\ y-4z=-7 \\ -4y+7z=10 \end{cases}$$

$$\begin{cases} x+y-z=2 \\ y-4z=-7 \\ -4y+7z=10 \end{cases} \xrightarrow{R_3+4R_2 \to R_3} \begin{cases} x+y-z=2 \\ y-4z=-7 \\ -9z=-18 \end{cases} \xrightarrow{-1/9\times R_3 \to R_3} \begin{cases} x+y-z=2 \\ y-4z=-7 \\ z=2 \end{cases}$$

By using backward substitution, we can solve y and then x finally:

Sub $z=2$ into R_2: $y-4(2)=-7 \quad \therefore\ y=1$

Sub $y=1$ and $z=2$ into R_1: $x+1-2=2 \quad \therefore\ x=3$

In practice, we will use **augmented matrix (增廣矩陣)** to represent the process in Gaussian Elimation:

$$\left(\begin{array}{ccc|c} 3 & 2 & 1 & 13 \\ 2 & 2 & -2 & 4 \\ 5 & 1 & 2 & 20 \end{array}\right) \xrightarrow{R_1 \leftrightarrow \frac{1}{2}R_2} \left(\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 3 & 2 & 1 & 13 \\ 5 & 1 & 2 & 20 \end{array}\right) \xrightarrow[R_3-5R_1 \to R_3]{R_2-3R_1 \to R_2} \left(\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & -1 & 4 & 7 \\ 0 & -4 & 7 & 10 \end{array}\right)$$

$$\xrightarrow{-R_2 \to R_2} \left(\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & 1 & -4 & -7 \\ 0 & -4 & 7 & 10 \end{array}\right) \xrightarrow{R_3+4R_2 \to R_3} \left(\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & 1 & -4 & -7 \\ 0 & 0 & -9 & -18 \end{array}\right) \xrightarrow{-\frac{1}{9}R_3 \to R_3} \left(\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & 1 & -4 & -7 \\ 0 & 0 & 1 & 2 \end{array}\right)$$

Finally, the 1st non-zero element in each row of $\left(\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & 1 & -4 & -7 \\ 0 & 0 & 1 & 2 \end{array}\right)$ is 1 and the number of unknowns in each row is always less than that in the previous row. The augmented matrix in this form is called **row echelon form** and it can be solved by backward substitution.

Not for sale



Gaussian Elimination

More Example:

Solve the system of equations $\begin{cases} 7x-5y-7z=-8 \\ 4x-2y-3z=0 \\ 5x+4y+4z=35 \end{cases}$.

Solution: By Gaussian Elimination,

$$\begin{pmatrix} 7 & -5 & -7 & -8 \\ 4 & -2 & -3 & 0 \\ 5 & 4 & 4 & 35 \end{pmatrix} \xrightarrow{1/7 \times R_1 \to R_1} \begin{pmatrix} 1 & -\frac{5}{7} & -1 & -\frac{8}{7} \\ 4 & -2 & -3 & 0 \\ 5 & 4 & 4 & 35 \end{pmatrix}$$

$$\xrightarrow[R_3 - 5R_1 \to R_3]{R_2 - 4R_1 \to R_2} \begin{pmatrix} 1 & -\frac{5}{7} & -1 & -\frac{8}{7} \\ 0 & \frac{6}{7} & 1 & \frac{32}{7} \\ 0 & \frac{53}{7} & 9 & \frac{285}{7} \end{pmatrix} \xrightarrow{7/6 \times R_2 \to R_2} \begin{pmatrix} 1 & -\frac{5}{7} & -1 & -\frac{8}{7} \\ 0 & 1 & \frac{7}{6} & \frac{16}{3} \\ 0 & 53 & 63 & 285 \end{pmatrix}$$

$$\xrightarrow{R_3 - 53R_2 \to R_3} \begin{pmatrix} 1 & -\frac{5}{7} & -1 & -\frac{8}{7} \\ 0 & 1 & \frac{7}{6} & \frac{16}{3} \\ 0 & 0 & \frac{7}{6} & \frac{7}{3} \end{pmatrix} \xrightarrow{6/7 \times R_3 \to R_3} \begin{pmatrix} 1 & -\frac{5}{7} & -1 & -\frac{8}{7} \\ 0 & 1 & \frac{7}{6} & \frac{16}{3} \\ 0 & 0 & 1 & 2 \end{pmatrix}$$

$\therefore$ The system is transformed to $\begin{cases} x-\frac{5}{7}y-z=-\frac{8}{7} \\ y+\frac{7}{6}z=\frac{16}{3} \\ z=2 \end{cases}$.

By back substitution, the unique solution is $x=3$, $y=3$ and $z=2$.

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Gaussian Elimination

Because not all systems of linear equations have a solution, the following situations can also be arised in Gaussian elimination:

Case 1: Consistent (有解) with unique solution (唯一解)

The SLE $\left(\begin{array}{ccc|c} 7 & -5 & -7 & -8 \\ 4 & -2 & -3 & 0 \\ 5 & 4 & 4 & 35 \end{array}\right)$ shown in last page has a unique solution

"$x=3,\quad y=3$ and $z=2$".

Case 2: Consistent (有解) with infinitely many solution (無限多的解)

Using $\begin{cases} x+2y+3z=4 \\ 2x+4y+6z=8 \\ x+y+z=6 \end{cases}$ as example, it can be rewritten as

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 1 & 1 & 1 & 6 \end{array}\right) \longrightarrow \left(\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 6 \end{array}\right) \longrightarrow \left(\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & 6 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

The equation R_3: 0 = 0 is ***redundant* (多餘)**, so the SLE consists of 3 unknowns in 2 equations, thus there are infinitely many solutions.

Case 3: Inconsistent / no solutions (無解)

Using $\begin{cases} x+2y+3z=4 \\ 2x+4y+6z=5 \\ x+y+z=6 \end{cases}$ as example, it can be rewritten as

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 5 \\ 1 & 1 & 1 & 6 \end{array}\right) \longrightarrow \left(\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & -3 \\ 1 & 1 & 1 & 6 \end{array}\right) \longrightarrow \left(\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & 6 \\ 0 & 0 & 0 & -3 \end{array}\right)$$

The equation R_3: 0 = –3 is inconsistence, so there are no solutions

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Gaussian Elimination

More Example:

Solve the system of equations $\begin{cases} x-y-3z=9 \\ -x-3y+11z=7 \\ x-13y+21z=53 \end{cases}$.

Solution:

$$\left(\begin{array}{cccc} 1 & -1 & -3 & 9 \\ -1 & -3 & 11 & 7 \\ 1 & -13 & 21 & 53 \end{array}\right) \sim \left(\begin{array}{cccc} 1 & -1 & -3 & 9 \\ 0 & -4 & 8 & 16 \\ 0 & -12 & 24 & 44 \end{array}\right) \sim \left(\begin{array}{cccc} 1 & -1 & -3 & 9 \\ 0 & -4 & 8 & 16 \\ 0 & 0 & 0 & -4 \end{array}\right)$$

The last equation is $0=-4$ which is inconsistence.
Hence, the system is inconsistent and there is no solution.

Note: It is important to identify which equation leads to the inconsistency.

More Example:

Solve the system of equations $\begin{cases} x+y-z=5 \\ 4x+3y-z=2 \\ 5x+3y+z=-11 \end{cases}$.

Solution:

$$\left(\begin{array}{cccc} 1 & 1 & -1 & 5 \\ 4 & 3 & -1 & 2 \\ 5 & 3 & 1 & -11 \end{array}\right) \sim \left(\begin{array}{cccc} 1 & 1 & -1 & 5 \\ 0 & -1 & 3 & -18 \\ 0 & -2 & 6 & -36 \end{array}\right) \sim \left(\begin{array}{cccc} 1 & 1 & -1 & 5 \\ 0 & -1 & 3 & -18 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

The given system is transformed to $\begin{cases} x+y-z=5 \\ -y+3z=-18 \end{cases}$

There are 3 unknowns but only 2 equations.
Hence, the system has infinitely many solutions.

By letting $z=t$, then by back substitution
$x=-2t-13,\quad y=3t+18,\quad z=t$ where t is constant.

When a system of equations has infinitely many solutions, we set one of the variables as t (it calls parameter 參數), and express the other two variables in terms of t.

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Gaussian Elimination

[M2 2015 Q.5]

Solve the following systems of linear equations in real variables x , y , z:

(a) $\begin{cases} x+y+z=2 \\ 2x+3y-3z=4 \end{cases}$;

(b) $\begin{cases} x+y+z=2 \\ 2x+3y-3z=4 \\ 3x+2y+kz=6 \end{cases}$, where k is a real constant. ***(6 marks)***

Gaussian Elimination

[M2 2013 Q.8]

Consider the following system of linear equations in x, y and z

$$(E)\quad \begin{cases} x - ay + z = 2 \\ 2x + (1-2a)y + (2-b)z = a+4, \text{ where } a,b \in \mathbf{R} \\ 3x + (1-3a)y + (3-ab)z = 4 \end{cases}$$

It is given that (E) has infinitely many solutions.

(a) Find the values of a and b.

(b) Solve (E). ***(5 marks)***

Gaussian Elimination

[M2 2021 Q.8]

Consider the following system of linear equations in x, y and z

$$(E)\quad \begin{cases} x + (d-1)y + (d+3)z = 4-d \\ 2x + (d+2)y - z = 2d-5, \text{ where } d \in \mathbf{R} \\ 3x + (d+4)y + 5z = 2 \end{cases}$$

It is given that (E) has infinitely many solutions.

(a) Find d. Hence, solve (E).

(b) Sonemone claims that (E) has a real solution (x, y, z) satisfying $xy + 2xz = 3$. Is the claim correct? Explain your answer.

(8 marks)



Homogeneous SLE (齊次方程組)

Homogeneous system of linear equations (SLE) refers to a system where all the constant terms on the right side of the equations are zero:

	Basic System of Equation (In the form $A\mathbf{x} = \mathbf{b}$)	Homogeneous System (In the form $A\mathbf{x} = \mathbf{0}$)
Example: (2×2 System)	$\begin{cases} 3x-2y=4 \\ x+y=3 \end{cases}$ On solving, $x = 2$ and $y = 1$	$\begin{cases} 3x-2y=0 \\ x+y=0 \end{cases}$ Obviously (顯而易見), $x = y = 0$
Example: (3×3 System)	$\begin{cases} 7x-5y-7z=-8 \\ 4x-2y-3z=0 \\ 5x+4y+4z=35 \end{cases}$ On solving, $x = 3$, $y = 3$ and $z = 2$	$\begin{cases} 7x-5y-7z=0 \\ 4x-2y-3z=0 \\ 5x+4y+4z=0 \end{cases}$ Obviously, $x = y = z = 0$

Homogeneous SLE must consist of **trivial solution (明顯解)**, which is **zero solution (零解)**, e.g. $x = y = z = 0$.

But there are some homogeneous SLE which consistof both zero solution and **non-zero solution!**

In $\begin{cases} x+2y+3z=0 \\ 2x+4y+6z=0 \\ x+y+z=0 \end{cases}$, the equation $2x+4y+6z=0$ is actually reduntant.

(i.e. $\left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 2 & 4 & 6 & 0 \\ 1 & 1 & 1 & 0 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$)

As a result, the above homogeneous SLE consists of:

- **trivial solution (明顯解)** , i.e. $x = y = z = 0$.
- **non-trivial solution (非明顯解)** , where $x = t$, $y = -2t$, $z = t$ where t is constant.



Homogeneous System of Equations

Theorem: Solution of a Homogeneous System of Equations
(齊次方程組的解)

A system of homogeneous equations $A\mathbf{x} = \mathbf{0}$ must be consistent.

Also, there are 2 possible cases for the solution of the system $A\mathbf{x} = \mathbf{0}$:

- If A is non-singular ($\det A \neq 0$), then the system has **trivial solution (zero solution) only**. (i.e., the system has unique solution only.)
- If A is singular ($\det A = 0$), then the system has **non-trivial solutions (non-zero solutions)**. (i.e., the system has infinitely many solutions.)

Example:

Suppose the system of linear equations

$$\begin{cases} x + y - \lambda z = 0 \\ x + \lambda y - z = 0 \\ \lambda x + y - z = 0 \end{cases}$$

has nontrivial solutions. Find all the values of λ.

Solution The system (*) has nontrivial solutions if

$$\begin{vmatrix} 1 & 1 & -\lambda \\ 1 & \lambda & -1 \\ \lambda & 1 & -1 \end{vmatrix} = 0$$

$$-\lambda - \lambda - \lambda + \lambda^3 + 1 + 1 = 0$$

$$\lambda^3 - 3\lambda + 2 = 0$$

$$(\lambda - 1)^2(\lambda + 2) = 0$$

Hence, $\lambda = -2$ or 1.



Homogeneous System of Equations



[P.Math 1999 Paper I Q.1]

Suppose the system of linear equations

$$(*): \begin{cases} x & +y & -\lambda z & = 0 \\ x & +\lambda y & -z & = 0 \\ \lambda x & +y & -z & = 0 \end{cases}$$

has nontrivial solutions.

(a) Find all the values of λ

(b) Solve (*) for each of the values of λ obtained in (a). ***(6 marks)***

HKDSE Questions (level 2)

1. ***[M2 2022 Q.8]***

Consider the following system of linear equations in x, y and z

$$(E)\quad \begin{cases} ax + 2y - z = 4k \\ -x + ay + 2z = 4 \\ 2x - y + az = k^2 \end{cases}, \text{ where } a, k \in \mathbf{R}$$

(a) Assume that (E) has a unique solution. Express y in terms of a and k.

(b) Assume that (E) has infinitely many solutions. Solve (E).

(7 marks)

HKDSE Questions (level 3)

2. [***M2 2019 Q.6***]

Consider the system of linear equations in real variables x, y and z

$$(E): \begin{cases} x - 2y - 2z = \beta \\ 5x + \alpha y + \alpha z = 5\beta \\ 7x + (\alpha - 3)y + (2\alpha + 1)z = 8\beta \end{cases}, \text{ where } \alpha, \beta \in \mathbf{R}.$$

(a) Assume that (E) has a unique solution.

(i) Find the range of values of α.

(ii) Express y in terms of α and β.

(b) Assume that $\alpha = -4$. If (E) is inconsistent, find the range of values of β.

(7 marks)

HKDSE Questions (level 3)

Not for sale

HKDSE Questions (level 3)

3. [***M2 2020 Q.11***]

Consider the system of linear equations in real variables x, y and z

$$(E): \begin{cases} x - y - 2z = 1 \\ x - 2y + hz = k \\ 4x + hy - 7z = 7 \end{cases}, \text{ where } h, k \in \mathbf{R}.$$

(a) (i) Assume that (E) has a unique solution.

(1) Prove that $h \neq -3$.

(2) Solve (E).

(ii) Assume that $h = -3$ and (E) is consistent.

(1) Prove that $k = -2$.

(2) Solve (E). ***(9 marks)***

(b) Consider the system of linear equations in real variables x, y and z

$$(F): \begin{cases} x - y - 2z = 1 \\ x - 2y + hz = -2 \\ 4x + hy - 7z = 7 \end{cases}, \text{ where } h \in \mathbf{R}.$$

Someone claims that there are at least two values of h such that (F) has a real solution (x, y, z) satisfying $3x^2 + 4y^2 - 7z^2 = 1$. Do you agree? Explain your answer. ***(4 marks)***

HKDSE Questions (level 3)

Not for sale

HKDSE Questions (level 4)

4. ***[M2 2016 Paper I Q.11]***

(a) Consider the system of linear equations in real variables x, y, z

$$(E): \begin{cases} x & +y & -z & = 3 \\ 4x & +6y & +az & = b \\ 5x & +(1-a)y & +(3a-1)z & = b-1 \end{cases}, \text{ where } a, b \in \mathbf{R} .$$

(i) Assume that (E) has a unique solution.

(1) Prove that $a \neq 2$ and $a \neq -12$.

(2) Solve (E).

(ii) Assume that $a = -2$ and (E) is consistent.

(1) Find b.

(2) Solve (E). ***(9 marks)***

(b) Is there a real solution of the system of linear equations

$$\begin{cases} x & +y & -z & = 3 \\ 2x & +3y & -z & = 7 \\ 5x & +3y & -7z & = 13 \end{cases} .$$

satisfying $x^2 + y^2 - 6z^2 > 14$? Explain your answer. ***(3 marks)***

HKDSE Questions (level 4)

Not for sale

HKDSE Questions (level 4)

5. [***M2 2018 Q.11]***

(a) Consider the system of linear equations in real variables x, y and z

$$(E): \begin{cases} x + ay + 4(a+1)z = 18 \\ 2x + (a-1)y + 2(a-1)z = 20, \\ x - y - 12z = b \end{cases}, \text{ where } a, b \in \mathbf{R}.$$

(i) Assume that (E) has a unique solution.

(1) Find the range of values of a.

(2) Solve (E).

(ii) Assume that $a = 3$ and (E) is consistent.

(1) Find b.

(2) Solve (E). ***(9 marks)***

(d) Consider the system of linear equations in real variables x, y and z

$$(E): \begin{cases} x + 3y + 16z = 18 \\ x + y + 2z = 10 \\ x - y - 12z = s \\ 2x - 5y - 45z = t \end{cases}, \text{ where } s, t \in \mathbf{R}.$$

Assume that (F) is consistent. Find s and t. ***(3 marks)***

HKDSE Questions (level 4)

Not for sale

HKAL Questions (level 5)

6. ***[Pure Math 1996 Paper I Q.9]***

Consider the system of linear equations

$$(*): \quad \begin{cases} x+2y-z=3 \\ x+y+2z=4 \end{cases}$$

(a) Solve (*). ***(3 marks)***

(b) Find the solutions of (*) that satisfy $xy+yz+zx=2$. ***(4 marks)***

(c) Find the possible values of a and λ ($a, \lambda \in \mathbf{R}$) so that

$$\begin{cases} x+2y-z=3 \\ x+y+2z=4 \\ ax+y+z=\lambda \end{cases}$$ is solvable. ***(4 marks)***

(d) Using (b), or otherwise, find all possible values of a and λ ($a, \lambda \in \mathbf{R}$) so that

$$\begin{cases} x+2y-z=3 \\ x+y+2z=4 \\ xy+yz+zx=2 \\ ax+y+z=\lambda \end{cases}$$ has at least one solution. ***(3 marks)***

HKAL Questions (level 5)

Not for sale

HKAL Questions (Level 5)

7. ***[Pure Math 2000 Paper I Q.8]***

Consider the system of linear equations

$$(S): \begin{cases} x & -y & -z & = a \\ 2x & +\lambda y & -2z & = b \\ x & +(2\lambda+3)y & +\lambda^2 z & = c \end{cases} \quad \text{where } \lambda \in \mathbf{R}\ .$$

(a) Show that (*S*) has a unique solution if and only if $\lambda \neq -2$.
Solve (*S*) for $\lambda = -1$. ***(7 marks)***

(b) Let $\lambda = -2$.

(i) Find the conditions on a , b and c so that (*S*) has infinitely many solutions.

(ii) Solve (*S*) when $a = -1$, $b = -2$ and $c = -3$. ***(4 marks)***

(c) Consider the system of linear equations

$$(T): \begin{cases} x & -y & -z & +3\mu & -5 & = 0 \\ 2x & -2y & -2z & +2\mu & -2 & = 0 \\ x & -y & +4z & -\mu & -1 & = 0 \end{cases} \quad \text{where } \mu \in \mathbf{R}.$$

Using the results in (b), or otherwise, solve (*T*). ***(4 marks)***

Not for sale

HKAL Questions (level 5)

Not for sale



More Questions in Past Papers

1. ***[Pure Math 1998 Paper I Q.8]***

Consider the system of linear equations

$$(E): \begin{cases} ax + y + bz = 1 \\ x + ay + bz = 1 \\ x + y + abz = b \end{cases}, \text{ where } a, b \in \mathbf{R}.$$

(a) Show that (*E*) has a unique solution if and only if $a \neq 2$, $a \neq 1$ and $b \neq 0$. Solve (*E*) in this case. ***(7 marks)***

(b) For each of the following cases, determine the value(s) of *b* for which (*E*) is consistent. Solve (*E*) in each case.

(i) $a = -2$,

(ii) $a = 1$. ***(6 marks)***

(c) Determine whether (*E*) is consistent or not for $b = 0$. ***(2 marks)***



More Questions in Past Papers

2. ***[Pure Math 2001 Paper I Q.9]***

Consider the system of linear equations

$$(S): \begin{cases} x + \lambda y + z = k \\ \lambda x - y + z = 1 \\ 3x + y + 2z = -1 \end{cases} \text{ where } \lambda, k \in \mathbf{R} .$$

(a) Show that (S) has a unique solution if and only if $\lambda \neq 0$ and $\lambda \neq 2$

(2 marks)

(b) For each of the following cases, determine the value(s) of k for which (S) is consistent. Solve (S) in each case.

(i) $\lambda \neq 0$ and $\lambda \neq 2$,

(ii) $\lambda = 0$,

(iii) $\lambda = 2$. ***(8 marks)***

(c) If some solution (x, y, z) of $\begin{cases} x + z = 0 \\ -y + z = 1 \\ 3x + y + 2z = -1 \end{cases}$

satisfies $(x-p)^2 + y^2 + z^2 = 1$, find the range of values of p.

(5 marks)



More Questions in Past Papers

3. ***[Pure Math 2005 Paper I Q.7]***

(a) Consider the system of linear equations in x, y, z

$$(E):\quad \begin{cases} x + ay + z = b \\ 2x + (a+3)y + (a-1)z = 0 \\ 3x + a^2y + (4a+1)z = -b \end{cases}, \text{ where } a, b \in \mathbf{R}.$$

(i) Find the range of values of a for which (E) has a unique solution. Solve (E) when (E) has a unique solution.

(ii) For each of the following cases, find the value(s) of b for which (E) is consistent, and solve (E) for such value(s) of b.

(1) $a = 1$,

(2) $a = -2$. ***(10 marks)***

(b) Suppose that a real solution of

$$\begin{cases} x - 2y + z = b \\ 2x + y - 3z = 0 \\ 3x + 4y - 7z = -b \end{cases}$$

satisfies $x^2 + y^2 + z^2 = b + 3$

where $b \in \mathbf{R}$. Find the range of values of b. ***(5 marks)***



More Questions in Past Papers

4. ***[Pure Math 2006 Paper I Q.7]***

Consider the system of linear equations in x, y, z

$$(E) =: \begin{cases} x & +ay & +z & =4 \\ x & +(2-a)y & +(3b-1)z & =3 \\ 2x & +(a+1)y & +(b+1)z & =7 \end{cases}, \text{ where } a, b \in \mathbf{R} .$$

(a) Prove that (E) has a unique solution if and only if $a \neq 1$ and $b \neq 0$.
Solve (E) in this case. ***(6 marks)***

(b) (i) For $a = 1$, find the value(s) of b for which (E) is consistent, and solve (E) for such value(s) of b.

(ii) Is there a real solution (x, y, z) of

$$\begin{cases} x + y + z = 4 \\ 2x + 2y + z = 6 \\ 4x + 4y + 3z = 14 \end{cases}$$

satisfying $x^2 - 2y^2 - z = 14$? Explain your answer.

(7 marks)

(c) Is (E) consistent for $b = 0$? Explain your answer. ***(2 marks)***



More Questions in Past Papers

5. ***[Pure Math 2003 Paper I Q.7]***

(a) Consider the system of linear equations in x, y, z

$$(E): \begin{cases} x & +ay & -z & = 0 \\ 2x & -y & +az & = -2a \\ -x & +2a^2y & +(a-3)z & = 2a \end{cases}, \text{ where } a \in \mathbf{R}.$$

(i) Find the range of values of a such that (E) has a unique solution. Solve (E) when (E) has a unique solution.

(ii) Solve (E) for

(1) $a = 1$,

(2) $a = -4$. ***(10 marks)***

(b) Suppose (x, y, z) satisfies

$$\begin{cases} x & +y & -z & = 0 \\ 2x & -y & +z & = -2 \\ -x & +2y & -2z & = 2 \end{cases}.$$

Find the least value of $24x^2 + 3y^2 + 2z$ and the corresponding values of x, y, z. ***(5 marks)***



More Questions in Past Papers

6. ***[Pure Math 2004 Paper I Q.7]***

(a) Consider the system of linear equations

$$(E): \begin{cases} x + (a-2)y + az = 1 \\ x + 2y + 4z = 1 \\ ax - y + 3z = b \end{cases}, \text{ where } a, b \in \mathbf{R}.$$

(i) Prove that (*E*) has a unique solution if and only if $a \neq 2$ and $a \neq 4$. Solve (*E*) in this case.

(ii) For each of the following case, determine the value(s) of *b* for which (*E*) is consistent, and solve (*E*) for such value(s) of *b*.

(1) $a = 2$,

(2) $a = 4$. ***(10 marks)***

(b) If all solution (x, y, z) of $\begin{cases} x + 2z = 1 \\ x + 2y + 4z = 1 \\ 2x - y + 3z = 2 \end{cases}$

satisfy $k(x^2 - 3) > yz$, find the range of values of *k*. ***(5 marks)***



More Questions in Past Papers

7. ***[Pure Math 2007 Paper I Q.7]***

(a) Consider the system of linear equations in x, y, z

$$\begin{cases} x - 3y = 1 \\ x + 5y + az = b, \text{ where } a, b \in \mathbf{R}. \\ 2x + ay - z = 2 \end{cases}$$

(i) Find the range of values of a for which (E) has a unique solution. Solve (E) when (E) has a unique solution.

(ii) Suppose that $a = -2$. Find the value(s) of b for which (E) is consistent, and solve (E) for such value(s) of b.

(8 marks)

(b) Is the system of linear equations

$$\begin{cases} x - 3y = 1 \\ x + 5y + z = 16 \\ 2x + y - z = 2 \\ x - y - z = 3 \end{cases}$$

consistent? Explain your answer. ***(3 marks)***

(c) Solve the system of linear equations

$$\begin{cases} x - 3y = 1 \\ x + 5y - 2z = 1 \\ 2x - 2y - z = 2 \\ x - y - z = 3 \end{cases}.$$

(4 marks)



More Questions in Past Papers

8. ***[Pure Math 2002 Paper I Q.8]***

(a) Consider the system of linear equations in x, y, z

$$(S):\quad \begin{cases} ax - 2y + z = 0 \\ x - y + 2z = b, \text{ where } a, b \in \mathbf{R}. \\ y + az = b \end{cases}$$

(i) Show that (S) has a unique solution if and only if $a^2 \neq 1$.
Solve (S) in this case.

(ii) For each of the following cases, determine the value(s) of b for which (S) is consistent, and solve (S) in such value(s) of b.
(I) $a = 1$,
(II) $a = -1$. ***(9 marks)***

(b) Consider the system of linear equations in x, y, z

$$(T):\quad \begin{cases} ax - 2y + z = 0 \\ x - y + 2z = -1 \\ y + az = -1 \\ 5x - 2y + z = a \end{cases}, \text{ where } a \in \mathbf{R}.$$

Find all the values of a for which (T) is consistent.
Solve (T) for each of these values of a. ***(6 marks)***



More Questions in Past Papers

9. ***[Pure Math 1999 Paper I Q.8]***

Consider the system of linear equations

$$(E): \begin{cases} x & +\lambda y & +z & = \lambda \\ 3x & -y & +(\lambda+2)z & = 7 \\ x & -y & +z & = 3 \end{cases} \text{ where } \lambda \in \mathbf{R} .$$

(a) Show that (E) has a unique solution if and only if $\lambda \neq \pm 1$. ***(3 marks)***

(b) Solve (E) for

(i) $\lambda \neq \pm 1$,

(ii) $\lambda = -1$,

(iii) $\lambda = 1$. ***(8 marks)***

(c) Find the conditions on a, b, c and d so that the system of linear equations

$$\begin{cases} x & +y & +z & = 1 \\ 3x & -y & +3z & = 7 \\ x & -y & +z & = 3 \\ ax & +by & +cz & = d \end{cases}$$

is consistent. ***(4 marks)***

More Questions in Past Papers

10. ***[Pure Math 1997 Paper I Q.8]***

Consider the following two system of linear equations:

$$(S): \begin{cases} (a+1)x + 2y - 2z = 0 \\ x + ay + 2z = 0 \\ 3x - y + (a-7)z = 0 \end{cases} \quad \text{and}$$

$$(T): \begin{cases} (a+1)x + 2y - 2z = 6 \\ x + ay + 2z = 5b - 1 \\ 3x - y + (a-7)z = 1 - b \end{cases} .$$

(a) If (S) has infinitely many solutions, find all the values of a.
Solve (S) for each of these values of a. ***(7 marks)***

(b) For the smallest value of a found in(a), find the value of b so that (T) is consistent.
Solve (T) for these values of a and b. ***(4 marks)***

(c) Solve the system of equations

$$\begin{cases} -x + 2y - 2\sqrt{z} = 6 \\ x - 2y + 2\sqrt{z} = -6 \\ 3x - y - 9\sqrt{z} = 2 \\ 3x - 4y - z = -11 \end{cases} .$$

(4 marks)

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